

Considerations in Diagnostic Testing

Diagnosis of disorders of genome instability or DNA repair is often a multistep process. A suggested sequence of steps includes clinical evaluation, cellular and biochemical assays, and molecular genetic testing. As new tests are developed and new information is obtained about these disorders, testing procedures may change accordingly. Decisions regarding the extent of testing should consider both cost and clinical utility—specifically, whether additional information would alter patient management or treatment.

For example, DNA sequencing is rarely required to diagnose xeroderma pigmentosum (XP) in a child with classical clinical features and documented cellular hypersensitivity to UV radiation, along with defective DNA repair. However, if a family has one child affected with XP and is considering having additional children, DNA sequencing may enable prenatal diagnosis through analysis of a trophoblast biopsy specimen.

Genetic counseling is an essential component of patient management for these inherited conditions. This role may be fulfilled by the treating physician or a trained genetic counselor.

In the United States, only laboratories certified under the Clinical Laboratory Improvement Amendments (CLIA) are permitted to perform diagnostic testing for patient care. Tests conducted in research laboratories typically have limited clinical applicability. However, a research laboratory may identify a disease-causing mutation that can subsequently be confirmed in a CLIA-certified laboratory for use in clinical decision-making. A list of laboratories offering testing for these disorders is available at:

<http://genetests.org>, a resource funded by the National Institutes of Health.

CHAPTER 110

Carcinogenesis: Chemical

Andrzej A. Dlugosz, Stuart H. Yuspa

Cutaneous Cancer and Public Health

Cutaneous cancers account for more than half of all malignancies diagnosed in the United States. In 2006, the American Cancer Society estimated over 1 million non-melanoma skin cancers (NMSCs), including 800,000 to 900,000 basal cell carcinomas (BCCs) and 200,000 to 300,000 squamous cell carcinomas (SCCs), as well as more than 62,000 cases of melanoma. The exact number of NMSCs is uncertain because, unlike other cancers, they are not routinely reported to cancer registries.

Although the majority of NMSCs are curable and associated with low mortality, a small proportion—primarily SCCs—can be fatal. Due to the high incidence of these tumors, NMSCs result in an estimated 1,000 to 2,000 deaths annually.